MACHINE LEARNING | 25CS2107E

Hospital Readmission Prediction

Using Elastic Net Regularized Logistic Regression

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Team 23 | Section 13 | A.Y. 2026-2027
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Background: why readmissions matter

A hospital readmission is when a patient returns for further treatment after discharge. Frequent readmissions raise costs, hurt care quality and strain hospital resources.

Domain

Machine Learning, Healthcare Analytics and Predictive Modeling, supporting clinical decision-making.

The model

Logistic Regression predicts readmitted or not. Elastic Net blends L1 and L2 penalties to cut overfitting and drop irrelevant features.

Why it is needed

Flag high-risk patients early, plan follow-up care, reduce preventable readmissions and lower healthcare costs.

Problem statement

Existing problem

- Hospitals struggle to identify patients likely to return after discharge
- Traditional methods miss high-risk patients
- Higher treatment costs and workload

Challenges

- Missing or incorrect data
- Imbalanced patient records
- Choosing important features
- Overfitting and prediction errors
- Patient data privacy

Who is affected

- Patients: extra treatment and expense
- Doctors: hard to spot high risk
- Hospitals: workload and cost
- Families: more caregiving
- Insurers: higher medical expense

Motivation

1 Why this project

Use machine learning to predict readmissions, spot high-risk patients early and apply Elastic Net to real healthcare data.

2 Real-world need

Early risk identification lets doctors plan follow-up, monitor recovery, cut avoidable visits and manage resources better.

3 Research motivation

Healthcare analytics is growing. Elastic Net selects features and limits overfitting, showing how ML can support clinical decisions.

Objectives

MAIN OBJECTIVE Build an Elastic Net Logistic Regression model that predicts hospital readmission risk and helps staff plan better follow-up care.

1

Data collection

Gather patient data; fix missing, duplicate and inconsistent values

2

Feature selection

Find the factors that drive readmission

3

Model development

Logistic Regression with Elastic Net regularization

4

Model evaluation

Accuracy, precision, recall, F1 and ROC-AUC

5

Healthcare planning

Use predictions to prioritise follow-up and treatment

Existing system vs proposed system

Existing system

Relies on clinical assessment, medical history, EHR records and basic statistical models.

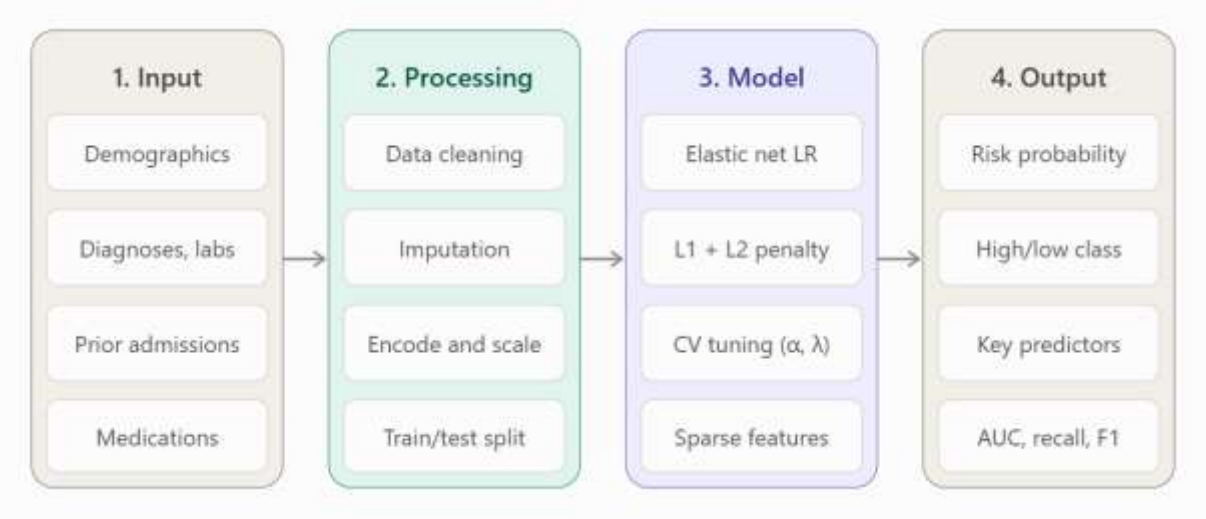
- Limited accuracy on high-risk patients
- Slow manual review of large data
- Hard to find the key factors
- Overfitting on new patients
- Varies across hospitals

Proposed system

Elastic Net Logistic Regression on patient data (age, history, prior visits) predicts whether a patient will be readmitted.

- Learns patterns tied to readmission
- Automates the analysis
- L1 picks key features
- L1 + L2 reduce overfitting
- Risk scores guide follow-up

System architecture



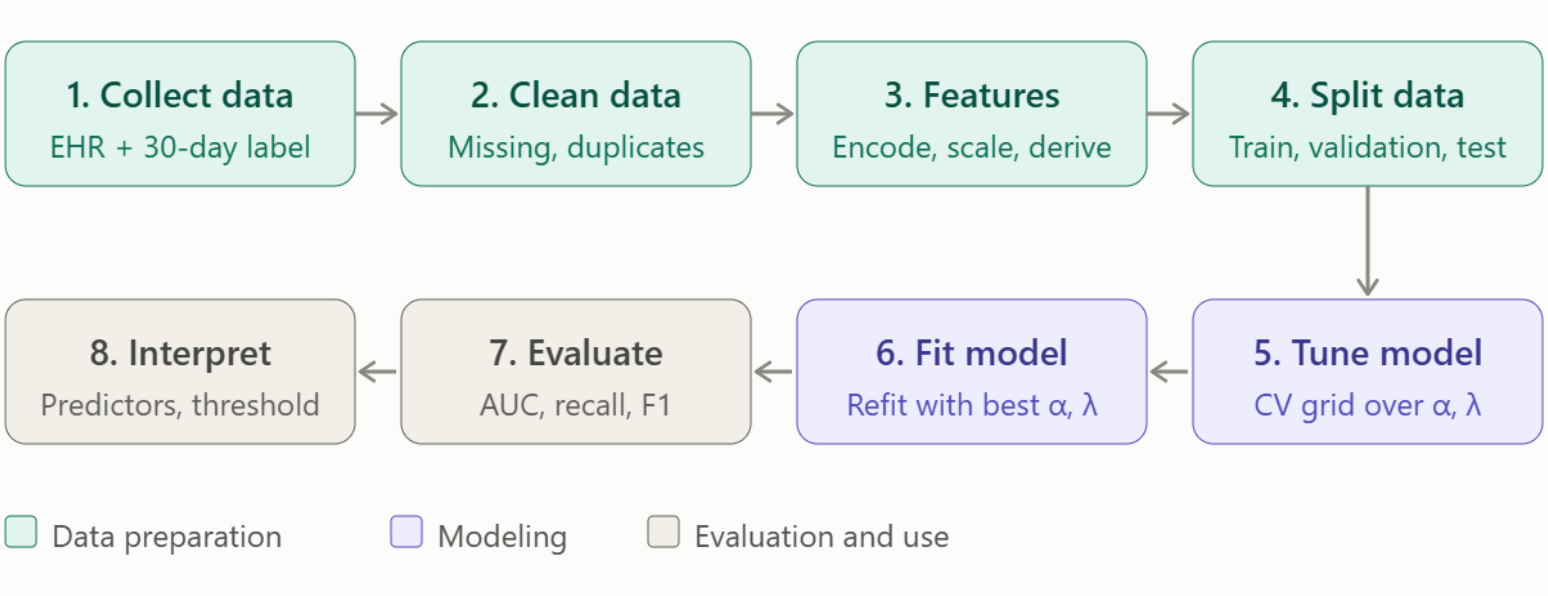
Input
EHR records and a 30-day readmission label

Processing
Clean, impute, encode, scale, split

Model
L1 + L2 logistic regression, CV tuned

Output
Probability, risk class, key predictors

Methodology and workflow



Imputation and scaling are fitted on the training set only to prevent leakage. Cross-validated AUC selects the mixing ratio and penalty strength.

Technology stack

Languages	Python 3.11, SQL, R (optional, glmnet check)
Frameworks	scikit-learn (LogisticRegression, elasticnet, saga); Flask / FastAPI; Streamlit
Libraries	pandas, NumPy, imbalanced-learn, SHAP / statsmodels, matplotlib, seaborn, joblib
Data	PostgreSQL / SQLite; MIMIC-IV or UCI Diabetes 130-US hospitals
Dev tools	Jupyter / VS Code, Git and GitHub
Hardware and cloud	Standard laptop (8-16 GB, no GPU); Colab, SageMaker or Azure ML; Docker

System modules

User interface

Enter or upload patient data, view risk scores

Data collection

EHR or public dataset via SQL, CSV, FHIR

Preprocessing

Impute, encode, scale and split

Core ML

Elastic Net LR with cross-validated tuning

Prediction

30-day probability and risk threshold

Visualisation

ROC, calibration, odds-ratio charts

Database

Raw and clean data, model versions, logs

Security

Role-based access, encryption, audit logs

Dataset

608	600	16	333 / 267
original records	after removing 8 duplicates	columns	readmitted / not

Hospital Readmission Dataset (Excel): demographics, admission details, medical history, clinical information and readmission status. Target: readmitted (1) or not (0). patient_id is excluded from training.

Patient

Age, gender

Admission

Type, diagnosis, length of stay, discharge disposition

History

Prior admissions, diabetes, hypertension, heart disease

Clinical

Medications, lab procedures, diagnoses

Other

Insurance type

Data preprocessing

1 Remove duplicates

8 duplicate rows dropped, leaving 600 records

2 Handle missing values

Median for age and lab procedures; most frequent for gender

3 Encode categories

One-hot encoding for gender, diagnosis, admission type, discharge, insurance

4 Scale features

Standardise numeric features to a common range

5 Split the data

Stratified 80/20: 480 training, 120 testing records

6 Prevent leakage

Target and patient ID kept out of the inputs

Algorithm: Elastic Net Logistic Regression



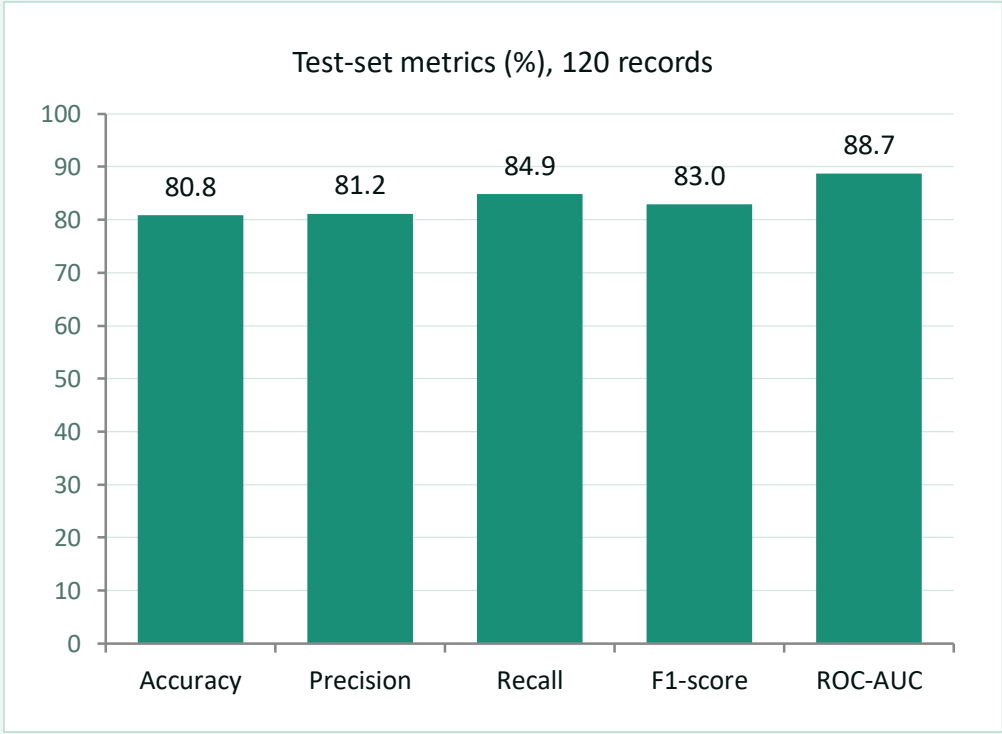
Why it fits

- Built for two outcomes: readmitted or not
- L1 shrinks weak features; L2 handles correlated ones
- Regularization limits overfitting on a small dataset
- Interpretable, with a probability for each patient

Parameters (preliminary run)

Penalty	**Elastic Net**
Solver	**SAGA**
C	**1.0**
l1_ratio	**0.5**
Max iterations	**5000**
Threshold	**0.5**
Random state	**42**

Results: model performance

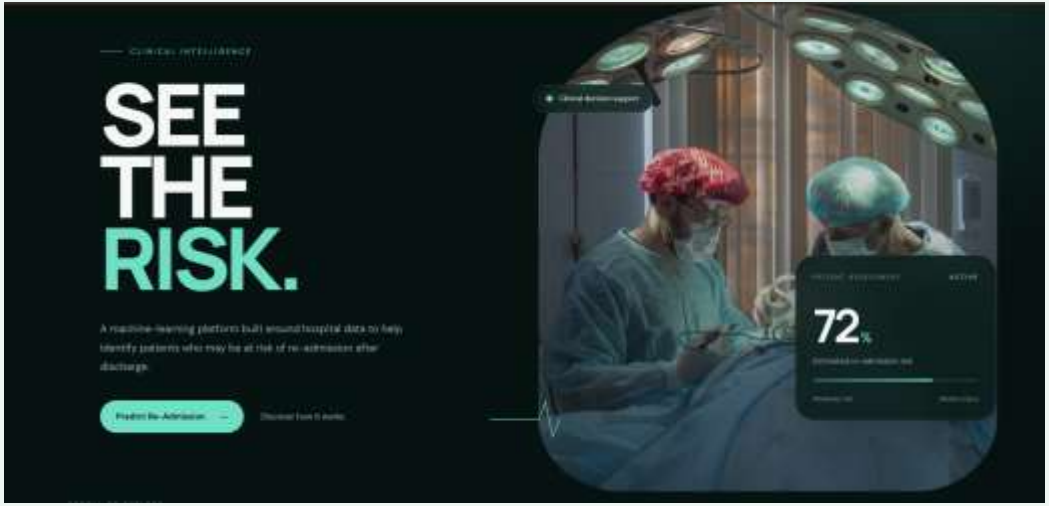


Confusion matrix

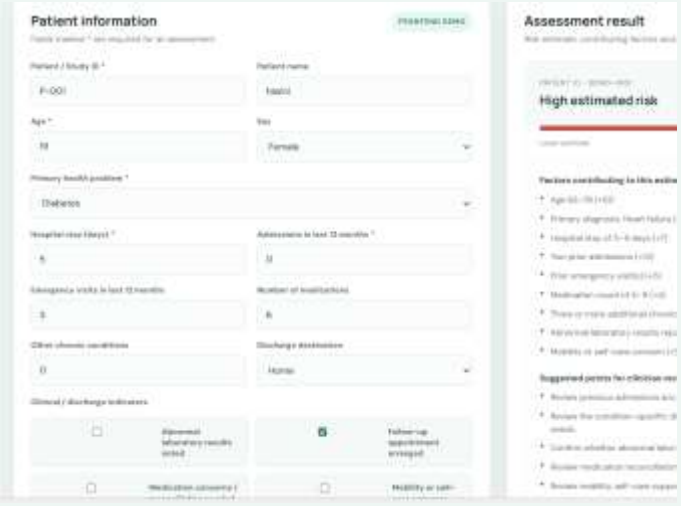
	Predicted: No	Predicted: Yes
Actual: No	**41** True negative	**13** False positive
Actual: Yes	**10** False negative	**56** True positive

Recall of about 85% means most real readmissions were caught. Preliminary result from a single 80/20 split, not clinical proof.

Demo: home page and patient input



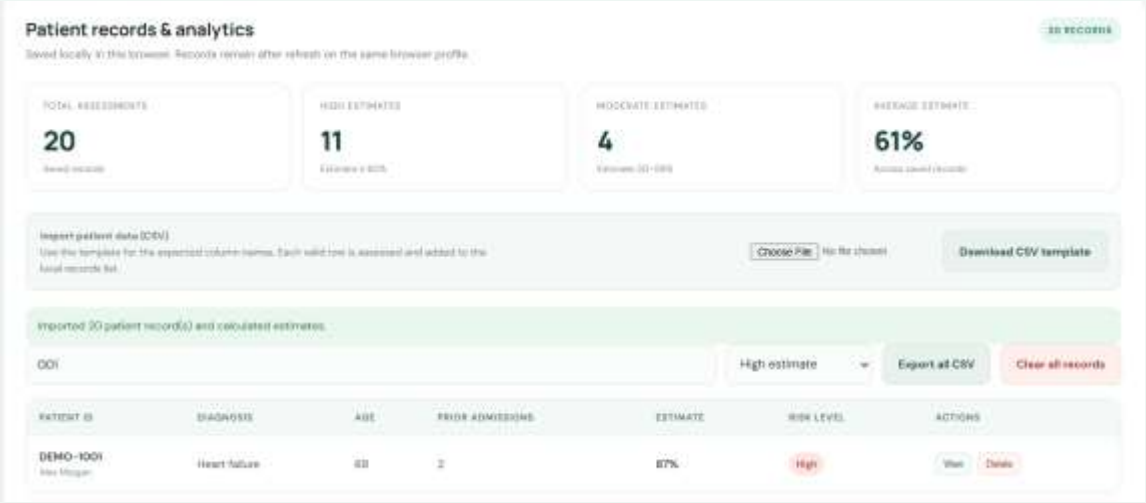
Home page



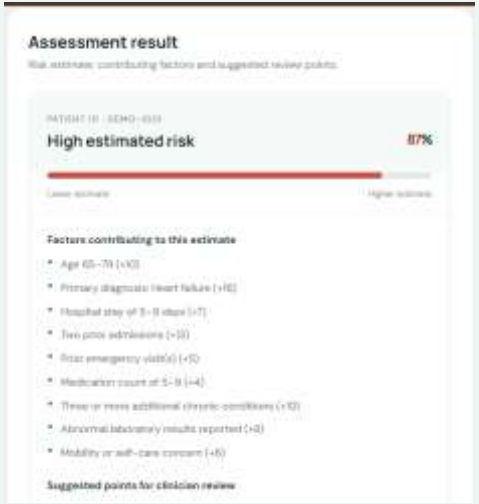
Input screen

Users enter patient details such as age, diagnosis, hospital stay, prior admissions and medications to request an assessment.

Demo: processing and risk result



Processing screen



Result screen

The result shows an estimated risk percentage, the factors contributing to it and suggested points for clinician review.

Advantages and applications

Advantages

- Early identification of at-risk patients
- Better follow-up and discharge planning
- Less manual analysis effort
- Elastic Net controls irrelevant and correlated features
- Interpretable, consistent predictions
- Supports hospital resource planning

Applications and users

- Hospitals and clinics: discharge planning
- Doctors: follow-up and monitoring decisions
- Administrators: beds, staff and resources
- Insurers: readmission and cost trends
- Researchers: better prediction models
- Meant to support, not replace, clinical judgment

Limitations and future scope

Limitations

- Prediction errors are possible
- May not generalise across hospitals
- Small dataset (600 records) and class imbalance
- Missing data and some clinical factors not captured
- Patient data privacy must be protected
- No comparison with other models yet

Future scope

- Larger, more diverse patient datasets
- Compare with standard LR and Decision Tree
- Cross-validation and parameter tuning
- Add medication, visit history and lifestyle factors
- Connect the web UI to a live prediction API
- Integrate with hospital systems at scale

Conclusion

Elastic Net Logistic Regression gives a simple, interpretable way to estimate hospital readmission risk.

81%	85%	0.89
accuracy on the test split	recall: most readmissions caught	ROC-AUC (88.69%)

The web interface lets users enter patient details, see estimated risk and contributing factors, and import or export CSV records. Next steps: connect a trained model, a secure database and real clinical data.

Thank you

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